

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$t_{sr(IN)}$ $t_{sf(IN)}$	Data input setup time	-	3	-	-	ns
$t_{hr(IN)}$ $t_{hf(IN)}$	Data input hold time	-	1	-	-	
$t_{vr(OUT)}$ $t_{vf(OUT)}$	Data output valid time	DHQC = 0	-	3.5	4.5	
		DHQC=1 All prescaler values (except 0)	-	$t_{CLK}/4 + 0.5$	$t_{CLK}/4 + 1.5$	
$t_{hr(OUT)}$ $t_{hf(OUT)}$	Data output hold time	DHQC = 0	1.5	-	-	
		DHQC=1, All prescaler values (except 0)	$t_{CLK}/4 - 0.5$	-	-	

1. DHQC must be set to reach the mentioned frequency.

Table 54. XSPI characteristics in DTR mode (with DQS) / HyperBus

Evaluated by characterization - Not tested in production.

All values apply to Octal- and Quad-SPI mode.

Delay block activated.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CLK}	Clock frequency	$2.7\text{ V} < V_{DD} < 3.6\text{ V}$ $C_L = 15\text{ pF}$	-	-	$72^{(1)(2)}$	MHz
$t_{w(CLKH)}$ $t_{w(CLKL)}$	OCTOSPI clock high and low time (even division)	$PRESCALER[7:0] = n$ $= (0, 1, 3, 5, \dots, 255)$	$t_{CLK}/2 - 0.5$ $t_{CLK}/2 - 0.5$	-	$t_{CLK}/2 + 0.5$ $t_{CLK}/2 + 0.5$	ns
$t_{w(CLKH)}$ $t_{w(CLKL)}$	OCTOSPI clock high and low time (odd division)	$PRESCALER[7:0] = n$ $= (2, 4, 6, \dots, 254)$	$(n/2) \times t_{CLK}/(n+1) - 0.5$ $(n/2+1) \times t_{CLK}/(n+1) - 0.5$	-	$(n/2) \times t_{CLK}/(n+1) + 0.5$ $(n/2+1) \times t_{CLK}/(n+1) + 0.5$	
$t_{v(CLK)}$	Clock valid time	-	-	-	$t_{CLK} + 2$	
$t_{h(CLK)}$	Clock hold time	-	$t_{CLK}/2 - 1$	-	-	
$t_{w(CS)}$	Chip select high time	-	$3 \times t_{CLK}$	-	-	ns
$t_{v(DQ)}$	Data input valid time	-	0	-	-	
$t_{v(DS)}$	Data strobe input valid time	-	0	-	-	
$t_{h(DS)}$	Data strobe input hold time	-	0	-	-	
$t_{v(RWDS)}$	Data strobe output valid time	-	-	-	$3 \times t_{CLK}$	
$t_{sr(DQ)}$ $t_{sf(DQ)}$	Data input setup time	-	-1 ⁽³⁾	-	-	
$t_{hr(DQ)}$ $t_{hf(DQ)}$	Data input hold time	-	3	-	-	
$t_{vr(OUT)}$ $t_{vf(OUT)}$	Data output valid time	DHQC = 0	-	3.5	4.5	
		DHQC = 1, All prescaler values (except 0)	-	$t_{CLK}/4 + 0.5$	$t_{CLK}/4 + 1.5$	
$t_{hr(OUT)}$ $t_{hf(OUT)}$	Data output hold time	DHQC = 0	1.5	-	-	
		DHQC = 1, All prescaler values (except 0)	$t_{CLK}/4 - 0.5$	-	-	

1. Maximum frequency values are given for a RWDS to DQ skew of maximum $\pm 1.0\text{ ns}$.